MILITARY SPECIFICATION

DECONTAMINATING APPARATUS: PORTABLE 14 LITER, M13, AND COMPONENTS FOR

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of assembled decontaminating apparatus and its major components (fluid container, pump assembly, hose assembly, brush, accessory container, and wand assembies). See 6.2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard: Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4230

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MILITARY

MIL-C-22750 - Coating, Epoxy-Polyamide.
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of.
MIL-D-50030 - Decontaminating Agent, DS2.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Ammunition Lot Numbering.
MIL-STD-45662 - Calibration Systems Requirements.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-51-527 - Decontaminating Apparatus: Portable, 14 Liter, M13.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

ASTM STANDARDS

B 117 - Salt Spray (Fog) Testing.
D 1654 - Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-51-527 and on the listed drawings shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the decontaminating apparatus shall conform to the specifications and drawings listed on Drawing 5-51-527.

3.2 Manufacture and assembly. The decontaminating apparatus shall be manufactured and assembled in accordance with Drawing 5-51-527.

3.3 Performance.

3.3.1 Fluid container (Drawing 5-51-538).

3.3.1.1 Lifting handle. The lifting handle shall support a load of at least 150 pounds for a minimum of 5 seconds without weld tears, or distortion of handle parts when tested in accordance with 4.4.5.1(a).

3.3.1.2 Burst strength. The fluid container shall not burst or rupture when subjected to a minimun internal hydrostatic pressure of 40 pounds per square inch gage (psig) for at least 5 minutes when tested in accordance with 4.4.5.1(b).

3.3.1.3 Leakage. The fluid container shall not leak when subjected to a minimum internal pressure of 15 psig, for a minimum of 15 seconds when tested in accordance with 4.4.5.1(c).

3.3.1.4 Plug assembly leakage (Drawing 5-51-578). The plug assembly for the fluid container shall not leak when subjected to 10 psig, for a minimum of 15 seconds when tested in accordance with 4.4.5.1(d).

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3.3.1.5 Drop strength. The fluid container shall not leak when dropped from a height of 4 feet onto solid concrete so as to strike on the weakest structural seam. The container shall be filled to 98 percent capacity with water, when tested in accordance with 4.4.5.1(e).

3.3.2 Pump (Drawing 5-51-540).

3.3.2.1 Physical strength. There shall be no leakage, deformation, or malfunction of the pump components and the pump shall conform to the Function requirement (3.3.2.2) after the pump piston has been extended horizontally and a bending load of at least 75 pounds has been attached to the end of the quick disconnect coupling for a period of 2 minutes when tested in accordance with 4.4.5.2(a).

3.3.2.2 Function.

(a) Pump vacuum. When the pump piston is fully extended from the fully closed position within 2 seconds, sufficient vacuum shall be developed in the chamber to maintain a vacuum of no less than 17 inches of mercury for a minimum of 30 seconds when tested in accordance with 4.4.5.2(b)(1).

(b) Pump pressure. When the pump piston is depressed from the fully extended position within 2 seconds, sufficient pressure shall be developed to maintain a pressure of at least 10 psig for a minimum of 30 seconds when tested in accordance with 4.4.5.2(b)(2).

(c) Pump efficiency. When the pump is operated at room temperature at no more than 35 strokes per minute at a 10-foot-suction-height, it shall empty a container filled with 14 liters of DS2 conforming to MIL-D-50030 (or approved DS2 simulant) with no more than 131 strokes of the pump piston when tested in accordance with 4.4.5.2(b)(3).

3.3.3 Hose assembly (Drawing 5-51-560).

3.3.3.1 Physical strength. The hose or coupling shall not break or the coupling separate from the hose when subjected to a static load of 150 pounds, for a minimum of 5 seconds when tested in accordance with 4.4.5.3(a).

3.3.3.2 Crush-resistance and recovery. The hose assembly shall yield no more than 50 percent of its unstressed diameter when a 225 pound static load has been applied over a 4 inch length of hose for a minimum of 2 minutes. When the load is removed the hose shall return to at least 98 percent of its unstressed diameter in not more than 2 minutes when tested in accordance with 4.3.5.1.

3.3.3.3 Air leakage. The hose and hose assembly shall not leak when subjected to 10 psig air pressure for a minimum of 15 seconds when tested in accordance with 4.4.5.3(b).

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3.3.3.4 Coil retention. After the hose assembly has been subjected to 160° ± 5°F for 12 hours, it shall retain its coiled straight line vertical height to within 25 percent of its original height when tested in accordance with 4.4.5.3(c).

3.3.4 Brush assembly (Drawing 5-51-561).

3.3.4.1 Bristle loss. The brush shall lose no more than 10 percent of its bristles when wetted with and in the presence of excess DS2 (conforming to MIL-D-50030) and used to scrub a surface with a minimum surface roughness value of a root-mean-square of 500, over a minimum distance of 12 inches, with a minimum of 15 pounds vertical force applied, at a minimum rate of 45 strokes per minute, for a continuous period of 10 minutes, and tested in accordance with 4.3.5.2.

3.3.4.2 Leakage. The brush shall not leak at wand/brush interface nor through brush block when subjected to a minimum water pressure of 20 psig for 30 seconds and tested in accordance with 4.4.5.5.

3.3.5 Decontaminating apparatus (Drawing 5-51-527) spray characteristics. At room conditions the decontaminating apparatus shall be capable of being primed with no more than 10 strokes of the pump piston. With the wand held horizontally a minimum of 4 feet above the outlet of fluid container, spray shall be projected a horizontal distance of at least 3 feet and no leakage shall occur when the pump is activated and tested in accordance with 4.4.5.4.

3.3.6 Accessory container assembly (Drawing 5-51-529).

3.3.6.1 Adhesion. The coating shall exhibit no removal from the container and the film shall show no blistering or other defects when tested as specified in 4.4.5.6(a).

3.3.6.2 Salt spray corrosion resistance. The container shall have a mean creepage from scribe of no more than 1.0 millimeter and shall show no evidence of rust or blisters on other concerned areas when tested as specified in 4.4.5.6(b).

3.4 Preproduction. Unless otherwise specified in the contract, a preproduction sample of decontaminating apparatus or the specified component shall be produced in accordance with this specification for examination and test (see 4.3 and 6.2), prior to the start of regular production.

3.5 Workmanship. The decontaminating apparatus and its components shall be free from cracks, dents, sharp edges, unpainted surfaces, rust, and foreign matter such as dirt, grease, and oily or viscous materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified

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in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lot. Unless otherwise specified (see 6.2) a preproduction lot of 100 M13 Decontaminating Apparatus (Drawing 5-51-527) and 10 of each of the following components:

(a) Fluid container assembly (Drawing 5-51-538)
(b) Pump assembly (Drawing 5-51-540)
(c) Hose assembly (Drawing 5-51-560)
(d) Brush (Drawing 5-51-561)
(e) Accessory container assembly (Drawing 5-51-529)

shall be manufactured using the same methods, processes, materials, and equipment that will be used during regular production.

4.3.2 Submission. Unless otherwise specified in the contract, the contractor shall submit a preproduction sample (4.3.1 and 4.3.3) for evaluation in accordance with the provisions of 4.3 and 6.2. The preproduction sample shall consist of the quantities and items specified in 4.3.1 and 6.2. A preproduction sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in

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excess of 90 days or whenever a change occurs in manufacturing process, material, drawings, specifications, or source of supply as to significantly affect product uniformity as determined by the Government.

4.3.3 Sampling.

4.3.3.1 For examination. A random sample of twenty M13 Decontaminating Apparatus and 10 of each component shall be taken from the preproduction lot for examination.

4.3.3.2 For nondestructive test. A random sample of twenty M13 Decontaminating Apparatus and all 10 of each component shall be taken from the preproduction lot for nondestructive testing.

4.3.3.3 For destructive test. A random sample of 10 fluid containers, 10 hose assemblies, 10 brushes, 3 accessory container assemblies (cleaned, pretreated and primed without topcoat) and 5 complete M13 decontaminating apparatus shall be taken from the preproduction lot for destructive test.

4.3.4 Inspection procedures.

4.3.4.1 For examination. The sample of M13 Decontaminating Apparatus shall be examined in accordance with the classification of defects. The Acceptable Quality Levels (AQLs) specified in the classification of defects do not apply. The acceptance number is zero. The sample of each component shall be examined for all the requirements specified on the applicable drawing. The acceptance number is zero.

4.3.4.2 For nondestructive tests. The sample of M13 Decontaminating Apparatus shall be tested for defects listed in the classification of defects. The AQLs do not apply; the acceptance number is zero. The sample component shall be tested, as applicable, for the defects listed in the classification of defects for that component. The AQLs do not apply; the acceptance number is zero.

4.3.4.3 For destructive test.

(a) Fluid containers. Five sample fluid containers shall be tested in accordance with 4.4.5.1(b) and five sample fluid containers shall be tested in accordance with 4.4.5.1(e); the acceptance number is zero.

(b) Hose assembly. Five hose assemblies from the sample shall be tested in accordance with 4.4.5.3(a); the acceptance number is zero. The remaining five hose assemblies from the sample shall be tested in accordance with 4.3.5.1; the acceptance number is zero.

(c) Brush. The 10 sample brushes shall be tested in accordance with 4.3.5.2; the acceptance number is zero.

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(d) Decontaminating apparatus spray characteristics. Five sample decontaminating apparatus shall be tested in accordance with 4.4.5.4; the acceptance number is zero.

(e) Accessory container assembly. Three sample container assemblies shall be tested in accordance with 4.4.5.6(b); the acceptance number is zero.

4.3.5 Tests. Inspection equipment which is to be acquired by the contractor and used in the performance of the following tests shall be in accordance with 4.4.4. Unless otherwise specified, tests shall be performed at room temperature, 20°-30°C (68°-86°F).

4.3.5.1 Hose assembly crush resistance and recovery. Measure the diameter of the hose. Uniformly apply the specified load to the specified length of hose (excluding couplings) for the required time. After the required period of time, measure the diameter of the stressed hose. Determine the percentage yield using the following formula:

$$Y = \frac{\text{Stressed diameter}}{\text{Original diameter}} \times 100$$

Remove the load and within the required time, measure the recovered diameter of the hose. Determine the percentage recovery using the following formula:

$$R = \frac{\text{Recovered diameter}}{\text{Original diameter}} \times 100$$

4.3.5.2 Brush assembly bristle loss. Weight brush so that a 15 pound vertical force will be applied when the brush is subjected to scrubbing action. Wet the brush and the test block surface with DS2 (maintain wetness on brush surface) (see 6.5). Scrub the specified length of surface at the required rate of strokes and for the specified time. Evaluate the bristle loss from brush.

4.3.6 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the preproduction sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements. The contractor shall receive written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of M13 Decontaminating Apparatus or specified component (see 6.2) shall consist of those like items produced at the same place utilizing the same batches of materials, lot of components, process runs, fabrication techniques, assembly methods, tools, equipment and facilities but shall not exceed one month's production. Each lot shall be controlled and identified in accordance with MIL-STD-1168. No more than one interfix lot number of filled fluid containers shall be contained in any one lot of M13 Decontaminating Apparatus.

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4.4.2 Sampling.

4.4.2.1 For examination. The M13 Decontaminating Apparatus or the specified component shall be sampled in accordance with MIL-STD-105 at the specified AQL (see 4.4.3.5).

4.4.2.2 For nondestructive test. The M13 Decontaminating Apparatus or the specified component shall be sampled in accordance with MIL-STD-105, level S-4 at the specified AQL (see 4.4.3.5). In addition, each fluid container assembly (Drawing 5-51-538) in the lot shall be subjected to leakage test.

4.4.2.3 For destructive test.

(a) Fluid container. The fluid container shall be sampled in accordance with MIL-STD-105, level S-1 at the specified AQL for both the burst strength (see 4.4.3.3(a)(1)) and the drop strength (see 4.4.3.3(a)(2)).

(b) Hose assembly. The hose assembly shall be sampled in accordance with MIL-STD-105, level S-1 at the specified AQL for physical strength (see 4.4.3.3(b)).

(c) Decontaminating apparatus. The decontaminating apparatus shall be sampled in accordance with MIL-STD-105, level S-1 at the specified AQL for spray characteristics (see 4.4.3.3(c)).

(d) Accessory container assembly. The accessory container assembly shall be sampled in accordance with MIL-STD-105, level S-1 at the specified AQL for salt spray corrosion resistance (see 4.4.3.3(d)).

4.4.3 Inspection procedures.

4.4.3.1 For examination. The sample decontaminating apparatus or specified component, and the level A packaging shall be examined in accordance with the classification of defects (4.4.3.5) and MIL-STD-105.

4.4.3.2 For nondestructive tests. The sample decontaminating apparatus or specified component shall be tested in accordance with the classification of defects (4.4.3.5) and MIL-STD-105. In addition, each fluid container assembly (Drawing 5-51-538) in the lot shall be tested in accordance with 4.4.5.1(c). Any container that leaks shall be rejected.

4.4.3.3 For destructive test.

(a) Fluid container.

(1) Burst strength. The sample fluid containers shall be tested in accordance with 4.4.5.1(b) at an AQL of 4.0 percent defective.

(2) Drop strength. The sample fluid containers shall be tested in accordance with 4.4.5.1(e) at an AQL of 4.0 percent defective.

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(b) Hose assembly physical strength. The sample hose assemblies shall be tested in accordance with 4.4.5.3(a) at an AQL of 2.5 percent defective.

(c) Decontaminating apparatus spray characteristics. Sample decontaminating apparatus shall be tested in accordance with 4.4.5.4 at an AQL of 2.5 percent defective.

(d) Accessory container assembly salt spray resistance. The sample accessory container assemblies shall be tested in accordance with 4.4.5.6(b) at an AQL of 2.5 percent defective.

4.4.3.4 For critical defects. Each fluid container in the lot shall be inspected for the critical defects listed in the classification of defects.

4.4.3.5 Classification of defects.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(a)	Decontaminating Apparatus: Portable, 14 Liter, M13				5-51-527
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width dimension incorrect		1.0%	3.1.2	CIE*
102	Height dimension incorrect		1.0%	3.1.2	CIE
103	Length dimension incorrect		1.0%	3.1.2	CIE
104	Components missing or incorrectly assembled		1.0%	3.1.2	Visual
105	Protective finish incorrect or missing		1.0%	3.1.2	CIE
Minor					
201	Labels missing or incorrectly assembled		2.5%	3.1.2	Visual
202	Workmanship		2.5%	3.5	Visual

NOTES:

CIE* - Commercial Inspection Equipment

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(b)	Fluid Container Assembly				5-51-538
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weldments incorrect		1.0%	3.1.2	Visual/CIE*
102	Height dimension incorrect		1.0%	3.1.2	CIE
103	Length, width and radius dimension incorrect		1.0%	3.1.2	CIE
104	Lifting handle strength (Note 1)	Note 2	0.65%	3.3.1.1	4.4.5.1(a)
105	Protective finish incorrect or missing		1.0%	3.1.2	CIE
Minor					
201	Components missing or incorrectly assembled		2.5%	3.1.2	Visual
202	Labels missing, incorrect or incorrectly located		2.5%	3.1.2	Visual
203	Workmanship		2.5%	3.5	Visual

NOTES: CIE* - Commercial Inspection Equipment
1. Alternate test; test shall be imposed either as above or in accordance with 4.4.3.5(n), category major 104.
2. Sample level S-4

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(c)	Accessory Container Assembly				5-51-529
					NEXT HIGHER ASSEMBLY
					5-51-527
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weldments incorrect		1.0%	3.1.2	CIE*/Visual
102	Components missing, incorrect or improperly assembled		1.0%	3.1.2	CIE
103	Perpendicularity incorrect		1.0%	3.1.2	CIE
104	Spring link lock location incorrect		1.0%	3.1.2	CIE
105	Clip bracket location incorrect		1.0%	3.1.2	CIE
106	Protective finish incorrect or missing		1.0%	3.1.2	CIE
107	Adhesion	Note 1	1.0%	3.3.6.1	4.4.5.6(a)
108	Overall height, overall width and overall depth incorrect		1.0%	3.1.2	CIE
109	Storage cover binding movement incorrect		1.0%	3.1.2	Functional
Minor					
201	Workmanship		2.5%	3.5	Visual
202	Label missing or incorrectly located		2.5%	3.1.2	Visual
203	Grease missing		2.5%	3.1.1	Visual

NOTES:

CIE* - Commercial Inspection Equipment

1. Sample level S-4

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(d)	Pump Assembly				5-51-540
					NEXT HIGHER ASSEMBLY
					5-51-527
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pump vacuum		100%	3.3.2.2(a)	4.4.5.2(b)(1)
102	Pump pressure		100%	3.3.2.2(b)	4.4.5.2(b)(2)
103	Physical strength	Note 1	0.65%	3.3.2.1	4.4.5.2(a)
104	Pump efficiency	Note 2	2.5%	3.3.2.2(c)	4.4.5.2(b)(3)
105	External components missing or incorrectly assembled		1.0%	3.1.2	Visual
106	Protective finish incorrect or missing		1.0%	3.1.2	CIE*
107	Nozzle cap thread incorrect (Dwg 5-51-545)		1.0%	3.1.2	CIE
108	Finish on areas specifying no finish		1.0%	3.5	Visual
Minor					
201	Workmanship		2.5%	3.5	Visual

NOTES:
1. Sample level S-4
2. Sample level S-1

CIE* - Commerical inspection equipment

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(e)	Saddle Assembly				5-51-546
					NEXT HIGHER ASSEMBLY 5-51-529
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside length incorrect		1.0%	3.1.2	CIE*
102	Weldments incorrect or improperly located		1.0%	3.1.2	CIE
103	Seam location incorrect		1.0%	3.1.2	CIE
104	Bottom and side not flush		1.0%	3.1.2	CIE
105	Inside width incorrect		1.0%	3.1.2	CIE
Minor					
201	Outside height incorrect		2.5%	3.1.2	CIE
202	Workmanship		2.5%	3.5	Visual

NOTES:

CIE* - Commercial Inspection Equipment

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(f)	Wand Assembly, Brush Half				5-51-537
					NEXT HIGHER ASSEMBLY
					5-51-527
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Components missing or incorrectly assembled		1.0%	3.1.2	Visual
102	Weldment incorrect		1.0%	3.1.2	CIE*/Visual
103	Protective finish incorrect or missing		1.0%	3.1.2	CIE
104	Adapter thread incorrect (Dwg 5-51-388)		1.0%	3.1.2	CIE
105	Wand union thread incorrect (Dwg 5-51-567)		1.0%	3.1.2	CIE
Minor					
201	Grease missing		2.5%	3.1.1	Visual
202	Workmanship		2.5%	3.5	Visual

NOTES:

CIE* - Commercial Inspection Equipment

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(g)	Wand Assembly, Pump Half				5-51-555
					NEXT HIGHER ASSEMBLY
					5-51-527
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Components missing or incorrectly assembled		1.0%	3.1.2	Visual
102	Weldment incorrect		1.0%	3.1.2	Visual/CIE*
103	Protective finish incorrect or missing		1.0%	3.1.2	CIE
104	Collar location incorrect		1.0%	3.1.2	CIE
105	Machine finish (32) incorrect (Dwg 5-51-553)		1.0%	3.1.2	CIE
106	Threads incorrect (Dwg 5-51-553)		1.0%	3.1.2	CIE
Minor					
201	Workmanship		2.5%	3.5	Visual
202	Grease missing		2.5%	3.1.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(h)	Brush				5-51-561
					NEXT HIGHER ASSEMBLY
					5-51-527
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Brush leakage		100%	3.3.4.2	4.4.5.5
102	Components missing or incorrectly assembled		1.0%	3.1.2	Visual
103	Length of bristles incorrect		1.0%	3.1.2	CIE*
104	Nozzle diameter incorrect		1.0%	3.1.2	CIE
105	Thread incorrect		1.0%	3.1.2	CIE
Minor					
201	Workmanship		2.5%	3.5	Visual
202	Flatness incorrect		2.5%	3.1.2	CIE
203	Scraper length incorrect		2.5%	3.1.2	CIE

NOTES:

CIE* - Commercial Inspection Equipment

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(i)	Hose Assembly				5-51-560
					NEXT HIGHER ASSEMBLY
					5-51-527
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Air leakage	Note 1	0.65%	3.3.3.3	4.4.5.3(b)
102	Coil retention	Note 1	0.65%	3.3.3.4	4.4.5.3(c)
103	Components incorrect or incorrectly assembled		1.0%	3.1.2	Visual
Minor					
201	Clamp location incorrect		2.5%	3.1.2	CIE*
202	Workmanship		2.5%	3.5	Visual

NOTES:

CIE* - Commercial Inspection Equipment

1. Sample level S-4

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(j)	Adapter Assembly				5-51-566
					NEXT HIGHER ASSEMBLY
					5-51-523
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Components missing, incorrect or incorrectly assembled		1.0%	3.1.2	Visual
102	Brazing incorrect		1.0%	3.1.2	Visual
Minor					
201	Workmanship		2.5%	3.5	Visual
202	Crimped end opening incorrect		2.5%	3.1.2	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(k)	Adapter Assembly				5-51-565
					NEXT HIGHER ASSEMBLY
					5-51-523
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Components missing, incorrect, or incorrectly assembled		1.0%	3.1.2	Visual
102	Length incorrect		1.0%	3.1.2	CIE*
103	Brazing incorrect		1.0%	3.1.2	CIE
Minor					
201	Workmanship		2.5%	3.5	Visual

NOTES:

CIE* - Commercial Inspection Equipment

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(1)	Plug Assembly				5-51-578
					NEXT HIGHER ASSEMBLY 5-51-538
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leakage	Note 1	0.65%	3.3.1.4	4.4.5.1(d)
102	Components missing, damaged or improperly assembled		1.0%	3.1.2	Visual

NOTES:

1. Sample level S-4

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 5-51-574 5-51-573, 5-51-582
4.4.3.5(m)	Packaging - Packing - Palletization				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Filler pads missing or incorrect		1.0%	5.1	Visual
102	Marking incorrect or missing		1.0%	5.3	Visual
103	Banding straps broken, missing or loose or incorrectly located		1.0%	5.4	Visual
104	Container size incorrect		1.0%	5.2	Visual
Minor					
201	Pallet incorrect, split or damaged		2.5%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.5(n)	Top-Reservoir Assembly				5-51-523
					NEXT HIGHER ASSEMBLY
					5-51-538
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lock clip location incorrect		1.0%	3.1.2	CIE*
102	Lifting handle location incorrect		1.0%	3.1.2	CIE
103	Weld or braze incorrect		1.0%	3.1.2	Visual
104	Lifting handle strength (Note 1)	Note 2	0.65%	3.3.1.1	4.4.5.1(a)
Minor					
201	Workmanship		2.5%	3.5	Visual

NOTES: CIE* - Commercial Inspection Equipment
1. Alternate test; test shall be imposed either as above or in accordance with 4.4.3.5(b), category Major 104.
2. Sample level S-4

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.4 Inspection equipment. Unless otherwise specified in the contract, the contractor shall furnish and maintain inspection equipment in accordance with MIL-I-45607 and MIL-STD-45662 (see 4.4.5 and 6.2).

4.4.5 Tests. Inspection equipment which is to be acquired by the contractor and used in the performance of the following tests shall be in accordance with 4.4.4. Unless otherwise specified, tests shall be performed at room temperature, 20°-30°C (68°-86°F).

4.4.5.1 Fluid container. The fluid container shall be tested as follows:

(a) Lifting handle. Secure the top reservoir or the container assembly and apply the specified static load uniformly over the handle for the specified time. Examine lifting handle and top-reservoir, or lifting handle and container assembly as applicable.

(b) Burst strength. Plug all openings to affect a seal within the fluid container. Attach a source of hydrostatic pressure into the container through a fitting and apply the specified hydrostatic pressure for the specified time. Note: The container sides may be supported in a fixture during this test. See 6.6 for precautions regarding high pressure.

(c) Leakage. With the fluid container openings sealed and with the specified air pressure introduced into the container through a fitting, submerge the container in water containing a wetting agent to a depth of no more than 3 inches below the surface for the specified time and observe for leakage. A steady stream of bubbles shall be an indication of leakage.

(d) Plug assembly leakage. Attach a source of air pressure through an airtight fitting to the open end of plug assembly and apply specified air pressure. Submerge assembly in water to a depth of no more than 3 inches and examine for leakage for the specified time. A steady stream of bubbles shall be an indication of leakage.

(e) Drop strength. Fill the container as specified and seal in accordance with note 4, Drawing 5-51-575. Drop the container as specified on bottom seam diagonally opposite from the vent/coupling area.

4.4.5.2 Pump assembly. The pump shall be tested as follows:

(a) Physical strength. Attach the pump in a holding fixture. Fully extend the piston. Apply the specified static load to the coupling on the end of the fully extended piston for the specified time. Examine for damage and perform the Function test.

(b) Function.

(1) Pump vacuum. Attach a vacuum gage at the coupling plug end and open the valve. Extend the pump piston from the fully closed position to the fully open position within the specified time and observe that the specified vacuum is maintained for the required time.

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(2) Pump pressure. Attach a pressure gage at the nozzle cap end. Fully extend the piston and depress piston within the specified time and observe that the required pressure is maintained for the specified time.

(3) Pump efficiency. Prepare a DS2 simulant solution of 50 percent glycerin and water. Fill the fluid container with no less than 14 liters of simulant and attach the hose and wand sections to the pump assembly. Position fluid container so that the nozzle is at the specified height above the container. Prime pump and pump simulant from the container at the specified rate and count and record the number of strokes required to empty the container. Thoroughly clean and dry pump.

4.4.5.3 Hose assembly. The hose shall be tested as follows:

(a) Physical strength. Secure one coupling of the hose and apply the specified load for the specified time to the other coupling.

(b) Air leakage. Plug one end of the hose assembly. Connect the other end to a controlled source of air pressure with a shut-off valve and a pressure gage. Assure that all connections are leak-free. Subject the hose assembly to the required pressure. Close the shut-off valve. Submerge assembly in water. Allow for the air in the hose assembly to reach equilibrium. Observe the pressure gage for the specified time. Any pressure drop or steady stream of bubbles shall indicate leakage (see 4.4.4).

(c) Coil retention. Suspend one end of the hose assembly by the adapter and allow the coil to extend by its own weight, measure the extended length. Recoil the hose assembly and subject the assembly to the specified temperature for the specified time. Remove the assembly and allow it to cool to room temperature. Repeat the extended length measurement and compare dimension with original length. The percent difference shall not exceed the specified limit.

4.4.5.4 Decontaminating apparatus spray characteristics. The decontaminating apparatus shall be tested as follows:

(a) Assemble the unit in accordance with the operating instructions (Drawing 5-51-571).

(b) Condition container to room temperature.

(c) With the valve open, operate the pump until fluid flows from the wand end. Record the number of strokes required.

(d) Hold the wand in specified position and operate pump and determine if pump projects a horizontal spray the required distance.

(e) Continue to operate for a minimum of five additional strokes and observe entire assembly for leakage.

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4.4.5.5 Brush assembly leakage. Assemble the brush to a suitable fixture and apply the specified water pressure for the specified time. Observe for leakage.

4.4.5.6 Accessory container assembly. The accessory container shall be tested as follows:

(a) Adhesion. The container shall be tested for coating adhesion in accordance with Adhesion (tape test) of MIL-C-22750 except that items to be tested shall be accessory container assemblies.

(b) Salt spray corrosion resistance. A straight scribe mark shall be made on the container as described in ASTM D 1654. The scribe mark shall not contact the edge of the specimen and must penetrate all coatings on the metal, leaving a uniformly bright line free of burrs. For the purpose of this test, the significant surface of the container shall be 3 inches on either side of the scribe mark. This significant area shall be from top to bottom of the container. Procedures for evaluation shall be in accordance with ASTM D 1654. Expose the cleaned, pretreated and primed (without topcoat) container to a 5 percent salt spray test for 336 hours in accordance with ASTM B 117. Upon removal, wash the container in warm running water maintained at a temperature of not greater than 100°F until free from any visible salt deposits and examine immediately in accordance with method 1 or 2 of ASTM D 1654. The mean creepage from scribe shall be as specified herein.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be levels A or C as specified (see 6.2).

5.1.1 Level A. The decontaminating apparatus shall be preserved and packaged in accordance with Drawing 5-51-573.

5.1.2 Level C. The decontaminating apparatus shall be unit packaged in accordance with Drawing 5-51-573 except the RSC fiberboard box and pads shall conform to type CF, class domestic, variety SW, grade 175 of PPP-B-636 and PPP-F-320 respectively. The box shall be closed in accordance with the closure requirements specified in the appendix of PPP-B-636 for domestic grade boxes.

5.2 Packing. Packing shall be levels A, B, or C as specified (see 6.2).

5.2.1 Level A. The decontaminating apparatus preserved and packaged as specified in 5.1.1 shall be packed in accordance with Drawing 5-51-582.

5.2.2 Level B. The decontaminating apparatus packaged as specified in 5.1.1 shall be packed in accordance with Drawing 5-51-582, except that the shipping container shall conform to class 1, style 1 of PPP-B-585.

5.2.3 Level C. The decontaminating apparatus, unit packed as specified in 5.1.1 or 5.1.2, shall require no further protection.

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5.3 Marking. In addition to any special marking required, the contract or order unit and shipping containers shall be marked in accordance with Drawing 5-51-573 and 5-51-582, as applicable. When required, the palletized unit load shall be marked in accordance with Drawing 5-51-574.

5.4 Palletized unit loads. When required, decontaminating apparatus packed in accordance with 5.2.1 and 5.2.2, shall be palletized on standard type, 4 way, 40 by 48 inch pallets in accordance with Drawing 5-51-574.

5.5 Component parts (hose, wands, brush, pump, fluid container, filled fluid container, and accessory container). When any of these items are being procured for storage and issue as repair parts, preservation, unit packaging, packing and marking shall be as specified on the respective Special Packaging Instructions (SPI).

6. NOTES

6.1 Intended use. The M13 Decontaminating Apparatus is a man-portable device intended for use in the partial decontamination of various combat vehicles and equipment.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether complete item or a major component is required.
(c) Level of preservation, packaging, packing, palletization required and shelf life code (see section 5).
(d) Preproduction/First Article:

(1) First Article Inspection and Report requirements.
(2) Delivery schedule for First Article Report.
(3) Destination and delivery schedule of First Article Samples when testing is performed by the Government.
(4) First Article Test report approval time.

(e) Production

(1) Destination and delivery schedule requirements.
(2) Production and inspection report requirements.

6.3 Submission of requests for inspection equipment approval. Contractor shall submit request for inspection equipment approval to AMSMC-QAV-T (A), Aberdeen Proving Ground, MD 21010-5423.

6.4 Submission of test results required by contract. Data shall be submitted in accordance with DD Form 1423 from the contractor.

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6.5 Sampling and testing precautions. DS2 is toxic and corrosive (alkaline) and should be handled with corrosive-resistant gloves, boots, and apron; an industrial faceshield or splash-proof goggles; full facepiece respirator with amine cartridge. For further information, see safety precautions on Drawing 5-51-502.

6.6 Caution. Due to the hazards inherent in testing with high pressures, contractors are advised to take all necessary precautions to protect personnel while testing.

6.7 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.8 Supersession data. This specification supersedes and includes the requirements of Chemical Research and Development Center Purchase Description EA-D-1288A, dated 16 November 1984.

6.9 Subject term (key word) listing.

Chemical agents
Decontamination
Decontamination equipment

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4230-A070

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-51533(EA)	DECONTAMINATING APPARATUS: PORTABLE 14 LITER, M13, AND COMPONENTS FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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